

In this study, we investigate the effectiveness of multimodal learning strategies for medical image analysis.

For benchmarking purposes, we first evaluate our models using two standard public datasets: MNIST and CIFAR-10.

These datasets serve only as preliminary baselines and are not central to our contribution.

The main contribution of this paper is the creation of a new dataset consisting of 3,200 chest X-ray images

collected from two regional hospitals in Portugal. The dataset includes patient metadata, radiologist

annotations, and diagnosis labels. We refer to this dataset as the HospitalX-CXR Collection throughout this paper.

Additional details about the collection protocol can be found in Silva et al. (2021), which is included in our references.

In addition to the HospitalX-CXR Collection, we also gathered a second dataset of 500 anonymized CT scans

from a partner clinic. The scans were manually segmented by two trained radiologists.

Although this dataset plays a crucial role in our evaluation, the authors do not formally name it

or provide any references or citations associated with its acquisition or ethical approval process.

We simply refer to it as “the CT scan dataset” in this paper.

References:

[1] Krizhevsky & Hinton. CIFAR-10 Dataset.

[2] Deng et al. MNIST Dataset.

[3] Silva et al. Protocols for Chest X-ray Acquisition in Regional Hospitals. 2021.